RKDF UNIVERSITY

Education Glorifies Nation

Syllabus of Ph. D. Entrance

Sub: Bio-Technology

Microbiology:

Classes of microorganisms and their characterization, nutrient requirements for growth; laboratory techniques in microbiology, pathogenic microorganisms and disease; applied microbiology; viruses, microbial genetics.

Immunology:

The origin of immunology; Inherent immunity; Humoral and cell mediated immunity; Primary and secondary lymphoid organ;; Antigen processing and presentation; Synthesis of antibody and secretion; Molecular basis of antibody diversity; Polyclonal and monoclonal antibody; Complement; Antigen-antibody reaction; Immune tolerance; Hyper sensitivity; Autoimmunity.

Biochemistry:

Biomolecules and their conformation; Chemical and functional nature of enzymes; Kinetics of enzyme catalyzed reactions; Bioenergetics; Metabolism of Carbohydrate, Protein and Lipid; Membrane transport and pumps; Cell cycle and cell growth control; Cell signaling and signal transduction; Biochemical and biophysical techniques for macromolecular analysis.

Basic Biotechnology and Bioprocess Engineering:

DNA; RNA; Replication; Transcription; Translation; Gene transfer Regulatory controls in prokaryotes and eukaryotes; recombination; Transposable elements, Molecular basis of genetic diseases and applications. Bioprocess technology for the production of cell biomass and primary/secondary metabolites, Microbial production, purification and bioprocess application(s) of industrial enzymes and Protein; Immobilization of enzymes and cells and their application for bioconversion processes.

Recombinant DNA Technology:

Restriction and modification enzymes; Vectors; cDNA and genomic DNA library; Hybridoma technology; Gene isolation; Gene cloning; Transposons and gene targeting; DNA labeling; DNA sequencing; Polymerase chain reactions; DNA fingerprinting; Southern and northern blotting; In-situ hybridization; RAPD; RFLP; Site-directed mutagenesis; Gene transfer technologies; Gene therapy; Genetic engineering in animal cell culture; Animal cell preservation.